

PROJECT REPORT

LegalEase

AI-Powered Legal Document Generator using Gemini

Project Type	AI / Generative AI Application
Technology	Gemini API, Web Application, Python/JavaScript
Academic Level	2nd B.Sc. Computer Science with AI
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Submitted as part of the Project Development Sessions

PHASE 1: BRAINSTORMING & IDEATION

Project Title: LegalEase – AI-Powered Legal Document Generator

1. Introduction

Brainstorming and Ideation is the first phase of the project development process. In this phase, a problem is identified and ideas are developed to create a suitable technological solution.

2. Problem Identification

Many people find it difficult to prepare legal documents because they may not know the correct format and legal terminology. Searching for document formats and organizing details can also take time.

3. Project Idea

LegalEase is a web-based application designed to help users generate basic legal document drafts by entering the required information through a simple and user-friendly interface. The generated draft should be reviewed before use.

4. Objectives

- To simplify the preparation of basic legal document drafts.
- To provide a user-friendly and easy-to-understand interface.
- To reduce the time required to organize document information.
- To generate structured draft content based on user input.
- To help users access basic document preparation support.

5. Proposed Solution

The proposed solution is to develop a web application where users select a document type, enter the required details, and generate a basic document draft. For example, a user can select a Rental Agreement and enter the necessary information.

6. Technologies Used

HTML is used to create the webpage structure, CSS is used for styling, and JavaScript is used to implement user interaction and basic draft generation.

7. Expected Outcome

The expected outcome is a simple prototype that demonstrates how LegalEase can collect user details and display a structured legal document draft for review.

8. Small Coding – Project Idea Display

```
<h1>LegalEase</h1>
<p>Generate basic legal document drafts easily.</p>
```

This code displays the project title and its purpose on a webpage.



PHASE 2: REQUIREMENT ANALYSIS

Project Title: LegalEase – AI-Powered Legal Document Generator

1. Introduction

Requirement analysis is the second phase of the project development process. In this phase, the requirements of LegalEase are identified and organized to clearly understand what the system should do and what is needed to develop it.

2. Problem Understanding

Users may find it difficult to prepare legal documents because they may not know the correct format, required details or legal terminology. LegalEase aims to simplify this process by collecting the necessary information through a simple interface and generating a structured draft.

3. User Requirements

- Users should be able to select the required type of legal document.
- Users should be able to enter the necessary personal and document details.
- Users should be able to generate a document draft easily.
- Users should be able to view and review the generated content before use.

4. Functional Requirements

- The system should provide a document-type selection option.
- The system should display suitable input fields based on the selected document.
- The system should process the entered information and generate a structured draft.
- The generated document should be displayed clearly for review.

5. Non-Functional Requirements

- The application should have a simple and user-friendly interface.
- The system should respond within a reasonable time.
- The application should be reliable and easy to maintain.
- User information should be handled securely.
- The generated output should be clear and properly formatted.

6. Hardware & Software Requirements

A computer or laptop with at least 4 GB RAM, a keyboard, mouse and internet connection are required for development and testing. HTML, CSS and JavaScript can be used to create the web interface and basic document generation. A modern web browser such as Chrome or Edge is required for testing.

7. Project Constraints

The project should be completed within the given academic timeline using the available resources and technologies. The generated legal documents are drafts and should be reviewed by an appropriate legal professional before real-world use.

8. Expected Outcome

The expected outcome of this phase is a clear requirement specification for LegalEase. It should define the users, required features, system requirements, technologies and constraints. These requirements will guide the Project Design and Development phases.

PHASE 3: PROJECT DESIGN PHASE

Project Title: LegalEase – AI-Powered Legal Document Generator

1. Introduction

The Project Design Phase defines how the LegalEase application will look and how its components will work together. The design is prepared based on the requirements identified in the previous phase.

2. Interface Design

The application is designed with a simple interface containing a document type selection area, input fields, a generate button and an output section for the generated document.

3. System Design

The system follows a simple flow: the user selects a document type, enters the required details, JavaScript validates the inputs and the application generates the document draft.

4. Data Flow

User details are collected through the form and passed to JavaScript functions. The processed information is then displayed in the document output section.

5. Design Features

- Simple navigation.
- Clear input fields.
- Easy-to-understand buttons.
- Structured document output.
- User-friendly layout.

6. Expected Outcome

The design phase provides a clear structure for implementing the LegalEase application and makes the development process easier.

7. Small Coding – Basic Design Structure

```
<div id="legalEase">
  <h2>LegalEase</h2>
  <input id="name" placeholder="Enter Name">
  <input id="address" placeholder="Enter Address">
  <button onclick="generateDocument()">
    Generate Document
  </button>
</div>
```



PHASE 4: PROJECT PLANNING PHASE

Project Title: LegalEase – AI-Powered Legal Document Generator

1. Introduction

The Project Planning Phase defines the activities, technologies and resources required to complete the LegalEase project successfully. It helps organize the development work in a systematic manner.

2. Project Activities

The major activities include requirement collection, interface design, coding, document generation, testing and final documentation.

3. Technology Planning

HTML and CSS are planned for the user interface, while JavaScript is planned for input validation and document generation. A web browser is used for testing the application.

4. Time Planning

The project is divided into different stages so that each phase can be completed systematically. Development and testing are performed after the requirements and design are finalized.

5. Resource Planning

The project requires a computer or laptop, a code editor, a web browser and the required programming knowledge. These resources are used to develop and test the application.

6. Risk Planning

Possible issues such as missing user inputs, incorrect formatting and JavaScript errors are considered during planning. Validation and testing are planned to reduce these problems.

7. Expected Outcome

The planning phase provides a clear development schedule and identifies the resources and activities required to complete the LegalEase project.

8. Small Coding – Project Flow

```
function projectFlow() {  
    console.log("Requirements");  
    console.log("Design");  
    console.log("Development");  
    console.log("Testing");  
    console.log("Final Output");  
}
```



PHASE 5: PROJECT DEVELOPMENT PHASE

Project Title: LegalEase – AI-Powered Legal Document Generator

1. Introduction

Project Development is the stage where the planned idea is converted into a working application. In this phase, the design, coding and implementation of the LegalEase system are carried out using web technologies.

2. Development Process

The LegalEase application is developed as a web-based system that helps users generate basic legal document drafts. The application allows users to enter required details, select a document type and generate a structured legal document.

3. User Interface Development

The user interface is developed using HTML and CSS. HTML is used to create forms, input fields, buttons and output sections. CSS is used to improve the appearance of the webpage and provide a simple and user-friendly design.

4. Functionality Implementation

JavaScript is used to implement the main functions of the project. It collects user information, validates input fields and generates legal document drafts. The system displays the generated document in a proper format.

5. Features Implemented

- User-friendly interface.
- Document type selection.
- User input validation.
- Automatic document generation.
- Display and print option for generated documents.

6. Technologies Used

- HTML
- CSS
- JavaScript

7. Expected Output

The system generates a structured legal document draft based on the user inputs and displays it on the screen.

8. Small Coding – Document Generation

```
function generateDocument() {
    let name = document.getElementById("name").value;
    let address = document.getElementById("address").value;
    let date = document.getElementById("date").value;

    if(name=="" || address=="" || date=="") {
        alert("Please fill all fields");
        return;
    }

    document.getElementById("output").innerText =
    "LEGAL DOCUMENT\n\n" +
    "Name: " + name + "\n" +
    "Address: " + address + "\n" +
    "Date: " + date;
}
```



PHASE 6: PROJECT TESTING PHASE

Project Title: LegalEase – AI-Powered Legal Document Generator

1. Introduction

Testing is an important phase of software development. It ensures that the LegalEase application works correctly and provides accurate results.

2. Testing Process

Different test cases are used to verify the functionality of the application. User inputs, document generation and validation features are tested carefully.

3. Input Validation Testing

The system checks whether all required fields are entered. If any field is empty, an error message is displayed.

4. Functional Testing

The document generation process is tested using different user inputs. The generated document is checked for proper formatting and correctness.

5. Error Testing

Invalid and incomplete inputs are tested to identify errors. Necessary corrections are made to improve the performance of the application.

6. Result

After testing, the LegalEase application works successfully and generates legal document drafts correctly.

7. Small Coding – Validation Testing

```
function validateForm() {  
    let name = document.getElementById("name").value;  
    let address = document.getElementById("address").value;  
  
    if(name=="" || address=="") {  
        alert("Please enter all details");  
        return false;  
    }  
  
    alert("Validation Successful");  
    return true;  
}
```



PHASE 7: PROJECT DOCUMENTATION

Project Title: LegalEase – AI-Powered Legal Document Generator

1. Introduction

Project Documentation explains the LegalEase project, its purpose, features, technologies, working process, and usage. It helps users and evaluators understand how the system works and how it can be maintained.

2. Project Overview

LegalEase is a web-based application that helps users generate basic legal document drafts by entering required information. The system organizes the entered details into a structured document format for review.

3. Objectives of Documentation

- * Explain the purpose, features, and workflow of the project.
- * Provide clear instructions for using the application.
- * Describe the technologies and main components used.
- * Record the project development and testing details.
- * Make the project easy to understand and maintain.

4. System Features

- * Selection of a basic legal document type.
- * User-friendly form for entering required details.
- * Generation of a structured legal document draft.
- * Simple and organized output for review.
- * Web-based interface accessible through a browser.

5. Technologies Used

HTML – webpage structure; **CSS** – styling and layout; **JavaScript** – user interaction and document generation.

6. User Guide

- * Open the LegalEase web application.
- * Select the required document type.
- * Enter the requested information in the form.
- * Submit the details and generate the document draft.
- * Review the generated output before using it.

7. Documentation Contents

The final documentation includes project overview, requirements, design details, development process, testing results, screenshots, usage instructions, limitations, future enhancements, and project references.

PHASE 8: PROJECT DEMONSTRATION

Project Title: LegalEase – AI-Powered Legal Document Generator

1. Introduction

The Project Demonstration phase presents the completed LegalEase application and explains its purpose, features, working process, and final output through a recorded demo video.

2. Demo Objective

The objective is to demonstrate that the application works as intended and clearly explain how a user can enter information and generate a structured legal document draft.

3. Demo Video Should Include

- * Project Name: LegalEase – AI-Powered Legal Document Generator.
- * Student introduction and brief project overview.
- * Purpose and problem addressed by the project.
- * Use and benefits of the application.
- * Screen sharing of the complete application.
- * Step-by-step execution and working process.
- * Final generated output.

4. Demonstration Flow

- * Open the LegalEase application.
- * Show the available document type or input form.
- * Enter sample user details.
- * Generate the legal document draft.
- * Display and explain the generated result.

5. Voice-over Explanation

During the demo, explain the project clearly. Describe what the user enters, what the application processes, and how the final document is produced.

6. Final Output

The demonstration should show a successful document generation process and the final structured legal document draft. The output should be readable and clearly visible in the recording.

7. Submission

Record the complete project demonstration, upload the video to Google Drive, and set the sharing permission to **Anyone with the link can view**. Submit the public Drive link along with the public GitHub repository link.